VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi-590018, Karnataka

A PROJECT REPORT ON

"AI-Powered KYC with Deepfake Detection & Behavioral Fraud Analysis"

Submitted in Partial fulfillment of the Requirements for the Degree of Bachelor of Engineering in Computer Science & Engineering
By

Nikhil Chaudhary (1CR23AI075)

Kirtan Shrestha(1CR23AI047)

Jenish Karki(1CR23AI042)

Nabin Khati(1CR23AI070)

Under the Guidance of,

Dr.Venkat

Asst Professor, Department of AIML, CMRIT

DEPARTMENT OF ARTIFICAL INTELLIGENCE AND MACHINE LEARNING

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI, BANGALORE-560037

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI, BANGALORE-560037

DEPARTMENT OF ARTIFICAL INTELLIGENCE AND MACHINE LEARNING

CERTIFICATE

Certified that the project work entitled

"AI-Powered KYC with Deepfake Detection & Behavioral Fraud Analysis"

is carried out by **Nikhil Chaudhary ,Kirtan Shresths,Jenish Karki,Nabin Khati** a bonafide students of CMR Institute of Technology in partial fulfillment for the award of Bachelor of Engineering in Artificial Intelligence and Machine Learning of the Visvesvaraya Technological University, Belgaum during the year 2024-2025. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Dr.Venkat	**Dr.Shyam P Joy**	**Dr.Sanjay Jain**
Project Guide	**Head of the Department**	**Principal**

ABSTRACT

Identity verification and KYC fraud prevention have become major challenges in today's digital world due to the rapid growth of online services, digital banking, and AI-generated identity fraud. The AI-Powered KYC Verification System is designed to automate document validation, face verification, and liveness detection, providing faster, smarter, and more secure identity authentication.

The system combines Artificial Intelligence, Machine Learning, and Computer Vision technologies into a unified platform. It uses DeepFace with ArcFace embeddings, RetinaFace detection, OpenCV preprocessing, and cosine similarity matching to compare a user's live face with the face extracted from uploaded identity documents. Liveness verification is also integrated to reduce spoofing attacks using photos or screens.

The architecture consists of a React.js frontend, a FastAPI backend, and an AI-based verification layer. Users can upload documents, complete live face verification, and receive approval or rejection results through an interactive dashboard. The backend handles OCR extraction, session management, face comparison, liveness checks, and database operations.

The project supports documents such as Aadhaar Card, PAN Card, Passport, and College ID cards. It demonstrates the practical application of AI and Computer Vision in secure digital identity verification and can be further extended with deepfake detection, biometric authentication, mobile deployment, and real-time fraud detection systems.

ACKNOWLEDGMENT

We take this opportunity to express our sincere gratitude and deep appreciation to our project guide for the valuable guidance, continuous support, and constant encouragement provided throughout the development of this project. The suggestions and feedback offered during each phase of the project greatly helped us in successfully completing the work.

We also extend our heartfelt thanks to the Head of the Department and all the faculty members of the Department of Artificial Intelligence and Machine Learning at CMR Institute of Technology for their motivation, support, and technical guidance during the course of this project.

We gratefully acknowledge the developers and contributors of the open-source tools and technologies used in this project, including FastAPI, React.js, DeepFace, OpenCV, and other Machine Learning libraries, whose contributions made the development of this AI-powered KYC Verification System possible.

Finally, we express our sincere gratitude to all researchers and academicians whose work in the fields of Artificial Intelligence, Computer Vision, Cybersecurity, and Digital Identity Verification inspired and guided us throughout the project development process.

Nikhil Chaudhary

Kirtan Shrestha

Jenish Karki

Nabin Khati

Contents

List of Tables

List of Figures

Chapter 1

INTRODUCTION

With the rapid growth of online services, digital banking, and fintech platforms, secure identity verification has become one of the most important challenges in today's digital world. Traditional Know Your Customer (KYC) systems are often manual, time-consuming, and vulnerable to identity fraud, fake documents, and spoofing attacks. The rise of AI-generated images and deepfake technologies has further increased the need for intelligent and secure verification systems capable of accurately validating user identities in real time.

The AI-Powered KYC Verification System is developed as a solution to these challenges. It is an intelligent identity verification platform that uses Artificial Intelligence, Machine Learning, and Computer Vision techniques to automate document verification, face matching, and liveness detection. The system is designed to provide secure, fast, and reliable digital identity authentication through a modern web-based interface and an AI-driven backend verification pipeline.

1. Background

Digital KYC verification has become an essential requirement in sectors such as banking, finance, e-commerce, healthcare, and online services. Existing systems often depend heavily on manual verification processes or basic image matching techniques, which may lead to delays, human errors, and security vulnerabilities. Additionally, many systems lack effective protection against spoofing attacks using printed photos, mobile screens, or manipulated facial images.

The advancement of Artificial Intelligence and Computer Vision technologies has enabled the development of smarter and more secure identity verification solutions. Face recognition models, OCR-based document extraction, and liveness detection techniques can now automate large parts of the KYC process while improving both accuracy and efficiency.

The AI-Powered KYC Verification System is built on this foundation. The system uses DeepFace with ArcFace embeddings, RetinaFace-based face detection, OpenCV preprocessing, and cosine similarity matching to compare a live webcam face with the face extracted from uploaded identity documents. A

React.js frontend and FastAPI backend work together to provide a complete end-to-end verification platform.

2. Problem Statement

Traditional KYC systems are often slow, manual, and vulnerable to identity fraud and spoofing attacks. Many systems fail to perform accurate face matching, secure liveness verification, or automated document validation. There is a need for an intelligent AI-based KYC system capable of securely verifying user identities using uploaded documents and live facial verification while minimizing false approvals and fraudulent attempts.

3. Objectives

The primary objectives of the AI-Powered KYC Verification System are as follows:

- Develop an AI-powered identity verification system using Machine Learning and Computer Vision techniques.
- Implement secure face verification using DeepFace and ArcFace embeddings.
- Support multiple identity documents including Aadhaar Card, PAN Card, Passport, and College ID cards.
- Perform OCR-based extraction and validation of document details.
- Integrate liveness verification to reduce spoofing attacks using photos or screens.
- Build a responsive and user-friendly dashboard using React.js for document upload and live face verification.
- Develop a FastAPI backend for API communication, face comparison, session handling, and database operations.
- Implement secure validation mechanisms for image quality, face extraction, and verification decisions.
-

4. Technologies Used

The AI-Powered KYC Verification System is developed using a modern technology stack that ensures scalability, security, and efficient processing. FastAPI is used as the backend framework for handling APIs and AI processing tasks, while React.js is used for building the frontend verification dashboard. DeepFace, ArcFace, RetinaFace, and OpenCV are used for face detection, facial embedding generation, image preprocessing, and similarity matching. MongoDB is used for storing verification sessions and results.

Component	Technology	Purpose
Frontend	React.js	Interactive KYC verification dashboard and user interface
Backend	FastAPI (Python)	RESTful API for handling verification requests, OCR processing, and face verification
AI Model	DeepFace + ArcFace	Core face recognition and facial similarity matching engine
Face Detection	RetinaFace + OpenCV	Face detection, image preprocessing, and face extraction
OCR Processing	EasyOCR / Tesseract OCR	Extraction and validation of document details
Database	MongoDB	Storage of verification sessions, results, and user data
Libraries	NumPy, OpenCV, Axios, Scikit-learn	Image processing, API communication, and AI operations
Security	Liveness Detection	Prevention of spoofing attacks using photos or screens
Version Control	Git and GitHub	Source code management and collaboration

Table 1.1: Technology Stack for AI-Powered KYC Verification System

Each technology was selected to provide high performance, accurate verification, and seamless integration between the frontend, backend, and AI-based verification pipeline.

5. Key Features and Advantages

The AI-Powered KYC Verification System offers several advanced features that make it more secure, intelligent, and efficient than traditional identity verification systems:

- **AI-Powered Face Verification:**
  The system uses DeepFace with ArcFace embeddings and cosine similarity matching to accurately compare a user's live face with the face extracted from uploaded identity documents.
- **Automated Document Verification:**
  The platform supports multiple identity documents such as Aadhaar Card, PAN Card, Passport, and College ID cards with automated OCR-based field extraction and validation.
- **Liveness Detection:**
  Integrated liveness verification helps prevent spoofing attacks using printed photos, mobile screens, or replayed facial images.
- **Modern Smart Dashboard:**
  The React.js frontend provides a clean and interactive multi-step verification interface for document upload, live face capture, and result visualization.
- **Secure Backend Architecture:**
  The FastAPI backend efficiently handles API communication, session management, face verification, OCR processing, and database operations.
- **Real-Time Verification Results:**
  Users receive verification status, similarity scores, liveness results, and confidence metrics instantly through the dashboard.
- **Scalable System Design:**
  The modular frontend-backend-AI architecture allows future expansion with features such as deepfake detection, biometric authentication, Aadhaar/PAN API integration, and mobile deployment.
- **Enhanced Fraud Prevention:**
  By combining OCR validation, face matching, and liveness checks, the system improves security and reduces the chances of identity fraud and fake verification attempts.

Chapter 2

LITERATURE REVIEW

The field of digital identity verification and KYC fraud prevention has gained significant importance due to the rapid growth of online banking, fintech platforms, e-commerce, and digital services. Advancements in Artificial Intelligence, Machine Learning, and Computer Vision have enabled the development of automated systems capable of performing secure and accurate identity verification.

Machine Learning for Face Verification

Machine Learning and Deep Learning techniques are widely used in modern face recognition systems. Earlier methods struggled with variations in lighting, pose, and image quality. Modern models such as FaceNet, ArcFace, and DeepFace provide higher accuracy by generating facial embeddings for reliable face matching. The AI-Powered KYC Verification System uses DeepFace with ArcFace embeddings and cosine similarity matching for secure verification.

AI-Based Identity Verification Systems

Digital identity verification systems are increasingly used in banking, healthcare, and online services. Traditional KYC methods are often manual, slow, and vulnerable to fraud. AI-powered systems automate document validation, face matching, and user authentication while improving both speed and security. This project combines OCR, face verification, and liveness detection into a unified platform.

Liveness Detection and Anti-Spoofing

Liveness detection helps prevent spoofing attacks using printed photos or replayed videos. Techniques such as texture analysis, motion detection, and deep learning-based anti-spoofing improve the reliability of face verification systems. The project integrates liveness verification to ensure that the captured face belongs to a real user.

OCR-Based Document Verification

OCR technologies are widely used for extracting information from identity documents such as Aadhaar Cards, PAN Cards, and Passports. OCR-based systems automate data extraction and reduce manual verification efforts. In this project, OCR is used to extract and validate document details before performing face verification.

Web-Based Verification Dashboards

Modern frontend frameworks such as React.js enable the development of interactive and responsive verification dashboards. The AI-Powered KYC Verification System uses React.js to provide a user-friendly interface for document upload, live face capture, and result visualization.

REST API Architectures for AI Systems

Machine Learning models are commonly deployed through REST APIs in modern AI applications. FastAPI is widely used because of its high performance and seamless integration with Python libraries. In this project, FastAPI acts as the backend layer connecting the React frontend with the AI-based verification pipeline.

Chapter 3

METHODOLOGY

The methodology adopted for the AI-Powered KYC Verification System follows a structured approach involving document processing, OCR extraction, face verification, liveness detection, and deployment.

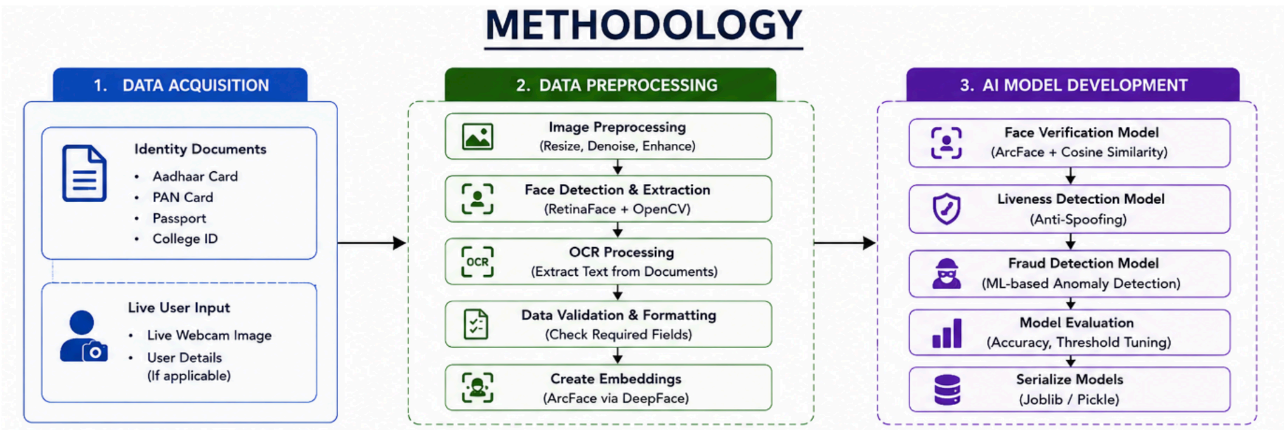

fig:Methodology

3.1 Dataset Description and Input Processing

The system uses uploaded identity documents such as Aadhaar Card, PAN Card, Passport, and College ID cards for verification. Live webcam images captured from users are also processed for real-time face verification and liveness detection. The system extracts important information such as name, date of birth, gender, and ID number from uploaded documents.

3.2 Data Preprocessing

Before verification, uploaded images undergo preprocessing steps such as image resizing, noise reduction, face extraction, and validation checks. OpenCV is used for image preprocessing, while RetinaFace is used for accurate face detection. OCR techniques are applied to extract document details for validation.

3.3 Face Verification Model Development

DeepFace with ArcFace embeddings is used as the primary face verification model. ArcFace generates facial embeddings that capture unique facial features, while cosine similarity matching is used to compare the face extracted from the document with the live webcam face.

The model offers several advantages:

- High face verification accuracy
- Efficient real-time processing
- Robustness against lighting and pose variations
- Reliable identity matching

The AI verification pipeline was implemented using Python libraries such as DeepFace, OpenCV, and NumPy.

3.4 System Architecture

The AI-Powered KYC Verification System is organized into three major layers:

Frontend Layer (React.js)

The frontend provides a responsive dashboard where users can upload documents, capture live webcam images, and view verification results. React.js was selected for its interactive and component-based architecture.

Backend Layer (FastAPI)

The FastAPI backend handles API communication, OCR processing, session management, liveness verification, face comparison, and database operations. It acts as the communication layer between the frontend and the AI verification pipeline.

AI Verification Layer

The AI layer consists of DeepFace, ArcFace, RetinaFace, and OpenCV modules responsible for face extraction, embedding generation, image preprocessing, and similarity matching. The verification result is then returned to the frontend dashboard.

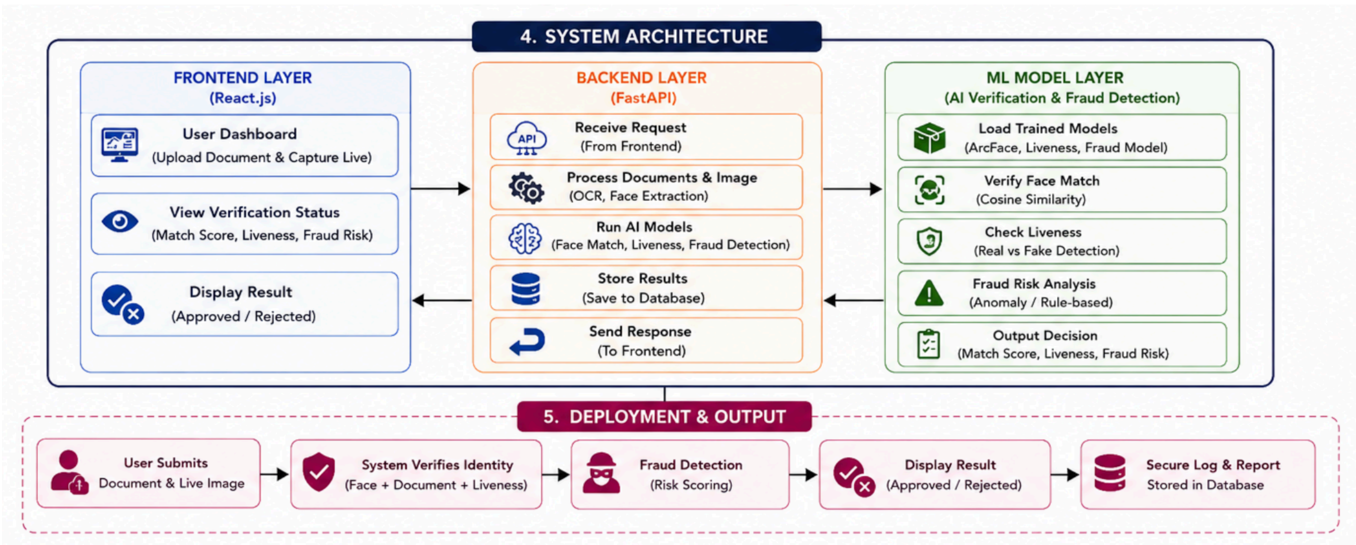

Fig:Workflow Diagram

3.5 User Workflow

The complete workflow of the AI-Powered KYC Verification System is as follows:

1. The user uploads an identity document through the dashboard.
2. The system extracts document details using OCR.
3. The face is extracted from the uploaded document.
4. The user captures a live webcam image for verification.
5. The AI model compares both facial embeddings using cosine similarity matching.
6. Liveness verification is performed to prevent spoofing attacks.
7. The system displays the final verification result as approved or rejected along with similarity and confidence scores.

Chapter 4

RESULTS AND DISCUSSION

The AI-Powered KYC Verification System was evaluated based on face verification accuracy, OCR extraction performance, liveness detection, and overall system functionality. This chapter presents the results obtained during testing and discusses their significance.

4.1 Face Verification Performance

The system was tested using multiple positive and negative verification cases. Positive tests involved matching the live webcam face with the correct identity document, while negative tests involved mismatched users and spoof attempts. The DeepFace model with ArcFace embeddings and cosine similarity matching produced reliable face verification results with high similarity scores for genuine users and lower scores for mismatched faces.

The system successfully performed:

- Face extraction from uploaded documents
- Live webcam face capture
- Facial embedding generation
- Cosine similarity matching
- Final approval or rejection decision

The similarity threshold mechanism helped distinguish genuine matches from fraudulent attempts, improving verification reliability.

4.2 OCR and Document Verification

OCR testing confirmed that the system could successfully extract important fields such as name, date of birth, gender, and ID number from uploaded documents including Aadhaar Cards, PAN Cards, Passports, and College ID cards. The extracted data was displayed correctly on the verification dashboard for validation.

4.3 Liveness Detection Results

The liveness verification module was tested against spoofing attempts using printed photos and mobile screens. The system successfully detected basic spoofing attempts by analyzing facial texture and image quality. This improved the security of the verification process by ensuring that the captured face belonged to a live user.

4.4 System Functionality

End-to-end testing confirmed that all components of the system functioned correctly together. The React.js frontend successfully handled document uploads, webcam capture, and result visualization. The FastAPI backend correctly processed API requests, performed OCR extraction, face verification, liveness checks, and returned verification results within acceptable response times.

The multi-step verification workflow was tested repeatedly, and the system consistently handled verification sessions without errors or state conflicts. The dashboard correctly displayed:

- Verification status
- Similarity scores
- Liveness results
- OCR extracted fields
- Final approval or rejection decisions

These results demonstrate that the AI-Powered KYC Verification System provides a reliable and scalable solution for secure digital identity verification using Artificial Intelligence and Computer Vision technologies.

Chapter 5

CONCLUSION

The AI-Powered KYC Verification System demonstrates the practical application of Artificial Intelligence, Machine Learning, and Computer Vision in secure digital identity verification. By combining DeepFace with ArcFace embeddings, OCR-based document verification, liveness detection, a React.js frontend, and a FastAPI backend, the project delivers a functional, scalable, and user-friendly identity verification platform.

The system successfully performs document upload, OCR extraction, face detection, live face verification, and approval or rejection decisions through an interactive dashboard. The use of cosine similarity matching and liveness verification improves the reliability and security of the verification process while helping reduce identity fraud and spoofing attacks.

The project follows a three-layer architecture consisting of the frontend, backend, and AI verification layer, ensuring proper separation of responsibilities and improving scalability and maintainability. The system also supports multiple identity documents such as Aadhaar Cards, PAN Cards, Passports, and College ID cards, making it suitable for various digital verification applications.

One of the major achievements of the project is the automation of traditional KYC verification processes, reducing manual effort and improving verification speed. The integration of OCR extraction, face matching, and liveness detection allows the system to perform secure identity verification in real time.

The project also provides a strong foundation for future enhancements. Features such as deepfake detection, advanced anti-spoofing models, Aadhaar/PAN API integration, biometric authentication, mobile application deployment, and blockchain-based identity storage can further improve system security and functionality.

In conclusion, the AI-Powered KYC Verification System proves that AI-driven technologies can be effectively used to build secure, intelligent, and scalable digital identity verification solutions for modern online platforms and services.

REFERENCES

1. C. S. Kubam, "Agentic AI Microservice Framework for Deepfake and Document Fraud Detection in KYC Pipelines," arXiv preprint arXiv:2601.06241, 2026.
2. F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A Unified Embedding for Face Recognition and Clustering," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2015.
3. Y. Dou, Z. Liu, L. Sun, Y. Deng, H. Peng, and P. S. Yu, "Enhancing Graph Neural Network-based Fraud Detectors against Camouflaged Fraudsters," in Proceedings of ACM CIKM, 2020.
4. Springer Neural Computing and Applications Journal Article on AI-based Fraud Detection Systems. Springer Article
5. J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019.
6. O. M. Parkhi, A. Vedaldi, and A. Zisserman, "Deep Face Recognition," in British Machine Vision Conference (BMVC), 2015.
7. A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," Communications of the ACM, vol. 60, no. 6, pp. 84–90, 2017.
8. A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in Advances in Neural Information Processing Systems (NeurIPS), 2019.
9. OpenCV Library Documentation, "Open Source Computer Vision Library," Available: https://opencv.org/
10. FastAPI Documentation, "FastAPI Framework for Building APIs with Python," Available: https://fastapi.tiangolo.com/
11. React.js Documentation, "React – A JavaScript Library for Building User Interfaces," Available: https://react.dev/
12. A. Geitgey, "Face Recognition Library," GitHub Repository, Available: https://github.com/ageitgey/face_recognition